Principles of communication systems

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Homework #05 Solutions

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Problem 1

$v(t) = A\cos(\theta(t)) \longrightarrow \omega_i = \frac{d\theta}{dt}$

a) $\omega_i = \omega$

b) $\omega_i = 2at + b$

c) $\omega_i = e^t$

d) $\omega_i = \omega\cos(\omega t + \theta)$

e) $\omega_i = \frac{1}{2\sqrt{t}}$

Problem 2

$$v(t) = A\cos\left(\int_{-\infty}^{t} \omega_i(\alpha)\,d\alpha\right)$$

a) $v(t) = A\cos\left(\omega_c t + \frac{1}{2}k_f t^2\right)$

b) $v(t) = A\cos\left(\omega_c t + \frac{1}{2}k_f C t\right)$

c) $v(t) = A\cos\left(\omega_c t + \frac{1}{\omega_m}\cos(\omega_m t)\right)$

d) $v(t) = A\cos\left(\omega_c t + k_f \int_{-\infty}^{t} u(\alpha - t_0)\,d\alpha\right)$

$$= A\cos\left(\omega_c t + k_f\, u(t - t_0)\, t\right)$$

Problem 3

$$m(t) = +1 \longrightarrow w_i = w_c + k_f$$

$$\varphi_{FM}(t) = A\cos\left((w_c + k_f)t\right), \quad T = \frac{2\pi}{w_c + k_f} = \frac{1}{f}$$

$$P_\varphi = \frac{1}{T}\int_{-T/2}^{T/2} |\varphi_{FM}(t)|^2 dt = \frac{A^2}{T} \cdot \int_{-T/2}^{T/2} \cos^2\left((w_c + k_f)t\right) dt$$

$$= \frac{A^2}{T}\left[\int_{-T/2}^{T/2} \frac{1}{2} \cdot dt + \int_{-T/2}^{T/2} \cos\left(2(w_c + k_f)t\right) dt\right] = \frac{A^2}{T} \cdot \frac{T}{2} = \frac{A^2}{2}$$

$m(t) = -1 \longrightarrow$ Similar

Problem 4

a) $V_{DC} = 0.6 \text{ V} \longrightarrow \mu = \dfrac{\text{Range}}{\text{Average}} = \dfrac{V_{PP}/2}{V_{DC}} = \dfrac{0.6}{0.6} = 1$

b) $f(t) = 0.6 + \underbrace{0.6}_{} \cos(2\pi(3\times10^3)t)$

1.2 peak to peak

c) $\varphi_{AM}(t) = f(t)\, A \cos(2\pi f_c t)$

$$= 0.6 A \cos(2\pi f_c t) + \frac{0.6 A}{2}\left[\cos(2\pi(f_c+f_m)t) + \cos(2\pi(f_c-f_m)t)\right]$$

$$= 6 \cos(2\pi(250\times10^3)t) + 3\left[\cos(2\pi(253\times10^3)t) + \cos(2\pi(247\times10^3)t)\right]$$

Problem 4

d) $P_s = \frac{3^2}{2} + \frac{3^2}{2} = 9$

$P_c = \frac{6^2}{2} = 18$

e) $\eta = \frac{P_s}{P_s + P_c} = \frac{9}{9+18} = 0.33 = 33\%$

f) $B =$ Max freq. $-$ Min freq

$= 253 \times 10^3 - 247 \times 10^3 = 6$ kHz

Problem 5

a) $m(t) = 0.6\cos(2\pi f_m t)$, $f_m = 3$ kHz

carrier $v(t) = 10\cos(2\pi f_c t)$, $f_c = 250$ kHz

$\varphi_{FM}(t) = B\cos(\theta(t))$, $\theta(t) = \omega_c t + k_f \int_{-\infty}^{t} m(\alpha)\,d\alpha$

$B = 10$

$\theta(t) = 2\pi f_c t + \frac{k_f(0.6)}{2\pi f_m}\sin(2\pi f_m t)$

b) According to problem 3, $P_\varphi = \frac{B^2}{2} = \frac{100}{2} = 50$

Problem 5

c) $$f_{max}[\varphi_{FM}] = f_c + \frac{k_f(0.6)}{2\pi} = 250 + 1.8 \quad kHz$$

$$f_{min}[\varphi_{FM}] = f_c - \frac{k_f(0.6)}{2\pi} = 250 - 1.8 \quad kHz$$

$$\beta = \frac{f_{max}[\varphi_{FM}] - f_{min}[\varphi_{FM}]}{3\ kHz} = \frac{3.6}{3} = 1.2$$

d) $$B_{FM} = 2(\beta + 1) f_m = 2(2.2)\,3\ kHz = 13.2\ kHz$$